INTRODUCTION TO COMPUTERS

1. **What is a Computer?**

An electronic device that stores, retrieves, and processes data, and can be programmed with instructions. A computer is composed of hardware and software, and can exist in a variety of sizes and configurations.

2. **Hardware & Software**

The term hardware refers to the physical components of your computer such as the system unit, mouse, keyboard, monitor etc.
The software is the instructions that makes the computer work. Software is held either on your computers hard disk, CD-ROM, DVD or on a diskette (floppy disk) and is loaded (i.e. copied) from the disk into the computers RAM (Random Access Memory), as and when required.

i) Hardware Components

Input Devices -- " How to tell it what to do"
-A keyboard and mouse are the standard way to interact with the computer. Other devices include joysticks and game pads used primarily for games.
Output Devices -- "How it shows you what it is doing"
-The monitor (the screen) is how the computer sends information back to you. A printer is also an output device.

INPUT DEVICES

The Mouse- Used to 'drive'Microsoft Windows

- The Keyboard - The keyboard is still the commonest way of entering information into a computer.
- Tracker Balls- an alternative to the traditional mouse and often used by graphic designers
- Scanners- A scanner allows you to scan printed material and convert it into a file format that may be used within the PC.
- Touch Pads- A device that lays on the desktop and responds to pressure.
- Light Pens- Used to allow users to point to areas on a screen .
- Joysticks- Many games require a joystick for the proper playing of the game.

OUTPUT DEVICES

- Monitor- The computer screen is used for outputting information in an understandable format
- Printers- There are many different types of printers. In large organizations laser printers are most commonly used due to the fact that they can print very fast and give a very high quality output.
- Plotters- A plotter is an output device similar to a printer, but normally allows you to print larger images.
- Speakers- Enhances the value of educational and presentation products.
- Speech synthesizers- Gives you the ability to not only to display text on a monitor but also to read the text to you

STORAGE DEVICES

"How it saves data and programs"

-Hard disk drives are an internal, higher capacity drive which also stores the operating system which runs when you power on the computer.
- disk drives allow you to save work on small disks and take the data with you.

Hard Disks
Speed: Very fast!
The speed of a hard disk is often quoted as "average access time" speed, measured in milliseconds. The smaller this number the faster the disk.
Capacity: Enormous! Often 40/80 Gigabytes. A Gigabyte is equivalent to 1024 Megabytes.

CD-ROM Disks
Speed: Much slower than hard disks. The original CD-ROM speciation is given a value of 1x speed, and later, faster CD-ROMs are quoted as a multiple of this value.
Capacity: Around 650 Mbytes and more

DVD Drives

Speed: Much faster than CD-ROM drives but not as fast as hard disks.
Capacity: Up to 17 Gbytes.

Memory
"How the processor stores and uses immediate data“
RAM -Random Access Memory
The main 'working' memory used by the computer.
When the operating system loads from disk when you first switch on the computer, it is copied into RAM.
As a rough rule, a Microsoft Windows based computer will operate faster if you install more RAM. Data and programs stored in RAM are volatile (i.e. the information is lost when you switch off the computer).

Memory

ROM –Read Only Memory
Read Only Memory (ROM) as the name suggests is a special type of memory chip that holds software that can be read but not written to.

- A good example is the ROM-BIOS chip, which contains read- only software.
- Often network cards and video cards also contain ROM chips.

Microprocessors
"The brain of the computer“ –
PCs primarily use microprocessors (sometimes called the chip). The older Intel versions include the 386, 486 and now the Pentium line.
The CPU (Central Processing Unit)is normally an Intel Pentium (or equivalent) and it is one of the most important components within your computer.

- It determines how fast your computer will run and is measured by its MHz speed.
- Thus a 600 MHz Pentium is much faster than say a 400 MHz Pentium CPU.
- It is the CPU that performs all the calculations within the computer.

TOPIC 2- INFORMATION TECHNOLOGY (IT)

Common Computer terms used on a day to day activities

Firewall - A hardware device or software to protect a computer from viruses, malware, trojans etc.

RAM (random-access memory) - any form of computer data storage that allow stored data to be accessed in any order (i.e., at random).
ROM (Read Only Memory) - a type of memory chip that retains its data when its power supply is switched off. □
Hard drive - a non-volatile storage device that stores data on rapidly rotating rigid (i.e. hard) platters with magnetic surfaces.

Hardware - the physical components of a computer.

Motherboard - the central printed circuit board (PCB) in many modern computers which holds many of the crucial components of the system, while providing connectors for other peripherals.

USB (Universal Serial Bus) - a specification to establish communication between devices and a host controller (usually a personal computers)

Virus - a computer program that can replicate itself and spread from one computer to another. The term "virus" is also commonly, but erroneously, used to refer to other types of malware, including but not limited to adware and spyware programs that do not have a reproductive ability.

Docker -Docker is an open-source program that enables a Linux application and its dependencies to be packaged as a container.

Translator- Translator is defined as a computer program that converts instructions written in one language to another without changing the initial logic in terms of computer language.

Rebooting - To reboot is to restart a computer and reload the operating system. On computers running Windows, you can usually reboot by selecting "turn off computer" from the start menu and then clicking "restart" in the window that pops up. Another way (and one that works sometimes when the first way doesn't) is through the Ctrl-Alt-Delete keystroke combination. (*warm boot,*)

Application software and System software

Applications software

An application program is the type of program that you use once the operating system has been loaded.

An application is a program, or group of programs, that is designed for the end user. software that allows users to do things like create text documents, play games, listen to music, or web browsers to surf the web are called application software .

Applications software (also called *end-user programs*) include such things as database programs, word processors, Web browsers and spreadsheets.

Application programs use the services of the computer's operating system and other supporting programs. The formal requests for services and means of communicating with other programs that a programmer uses in writing an application program is called the application program interface (API).

Figuratively speaking, applications software sits on top of systems software because it is unable to run without the operating system and system utilities.

1) Opera (Web Browser)
2) Microsoft Word (Word Processing)
3) Microsoft Excel (Spreadsheet software)
5) MySQL (Database Software)
6) Microsoft Powerpoint (Presentation Software)
7) iTunes (Music / Sound Software)
8) VLC Media Player (Audio / Video Software)
9) World of Warcraft (Game Software)
10) Adobe Illustrator (Graphics Software)

Operating systems software
The operating system is a special type of program that loads automatically when you start your computer.

The operating system allows you to use the advanced features of a modern computer without having to learn all the details of how the hardware works
The link between the hardware and you, the user.
Makes the computer easy to use without having to understand bits and bytes!

In contrast, Systems software consists of low-level programs that interact with the computer at a very basic level. This includes operating systems, compilers, and utilities for managing computer resources.

System software

System software is a type of computer program that is designed to run a computer's hardware and application programs. If we think of the computer system as a layered model, the system software is the interface between the hardware and user applications.

The operating system (OS) is the best-known example of system software.

The OS manages all the other programs in a computer.

Examples :
1) Microsoft Windows
2) Linux
3) Unix
4) Mac OSX
5) DOS
6) BIOS Software
7) HD Sector Boot Software
8) Device Driver Software i.e Graphics Driver etc
9) Linker Software
10) Assembler and Compiler Software

Other examples of system software and what each does:

- The BIOS (basic input/output system) gets the computer system started after you turn it on and manages the data flow between the operating system and attached devices such as the hard disk, video adapter, keyboard, mouse, and printer.
- The boot program loads the operating system into the computer's main memory or random access memory (RAM).
- An assembler takes basic computer instructions and converts them into a pattern of bits that the computer's processor can use to perform its basic operations.
- A device driver controls a particular type of device that is attached to your computer, such as a keyboard or a mouse. The driver program converts the more general input/output instructions of the operating system to messages that the device type can understand.
- According to some definitions, system software also includes system utilities, such as the disk defragmenter and System Restore, and development tools such as compilers and debuggers.

System software and application programs are the two main types of computer software. Unlike system software, an application program (often just called an application or app) performs a particular function for the user. Examples (among many possibilities) include browsers, email clients, word processors and spreadsheets.

OS- operating system

What is an operating system?

The operating system (prominent examples being, Microsoft Windows, Mac OS X and Linux), allows the parts of a computer to work together by performing tasks like transferring data between memory and disks or rendering output onto a display device.

It also provides a platform to run high-level system software and application software.

Operating system is the most important software that runs on a computer. It manages the computer's memory, processes, and all of its software and hardware. It also allows you to communicate with the computer without knowing how to speak the computer's language. Without an operating system, a computer is useless.

The operating system's job

Your computer's operating system (OS) manages all of the software and hardware on the computer.

Most of the time, there are many different computer programs running at the same time, and they all need to access your computer's central processing unit (CPU), memory, and storage.

The operating system coordinates all of this to make sure each program gets what it needs.

Types of operating systems

Operating systems usually come preloaded on any computer you buy.

Most people use the operating system that comes with their computer, but it's possible to upgrade or even change operating systems.

The three most common operating systems for personal computers are Microsoft Windows, Apple Mac OS X, and Linux.

Modern operating systems use a **graphical user interface**, or **GUI** (pronounced **gooey**). A GUI lets you use your mouse to click **icons**, **buttons**, and **menus**, and everything is clearly displayed on the screen using a combination of **graphics** and **text**.

Each operating system's GUI has a different look and feel, so if you switch to a different operating system it may seem unfamiliar at first. However, modern operating systems are designed to be **easy to use**, and most of the basic principles are the same.

CUI and **GUI** are two techniques used to send commands to a computer processor. Both accomplish the same goal, but employ somewhat different methods to do so.

CUI vs GUI

CUI and GUI are acronyms that stand for different kinds of user interface systems. These are terms used in reference to computers.

CUI stands for Character User Interface while

GUI refers to Graphical User Interface. Though both are interfaces and serve the purpose of running the programs, they differ in their features and the control they provide to the user.

Here is a brief explanation of the two types of user interface for the help of those who do not know about them.

What is CUI?

CUI means you have to take help of a keyboard to type commands to interact with the computer. You can only type text to give commands to the computer as in MS DOS or command prompt.

There are no images or graphics on the screen and it is a primitive type of interface.

In the beginning, computers had to be operated through this interface and users who have seen it say that they had to contend with a black screen with white text only.

In those days, there was no need of a mouse as CUI did not support the use of pointer devices. CUI's have gradually become outdated with the more advanced GUI taking their place.

However, even the most modern computers have a modified version of CUI called CLI (Command Line Interface).

What is GUI?

GUI is what most modern computers make use of. This is an interface that makes use of graphics, images and other visual clues such as icons.

This interface made it possible for a mouse to be used with a computer and interaction really became very easy as the user could interact with just a click of the mouse rather than having to type every time to give commands to the computer.

Difference between CUI and GUI

• CUI and GUI are user interface used in connection with computers

• CUI is the precursor of GUI and stands for character user interface where user has to type on keyboard to proceed. On the other hand GUI stands for Graphical User Interface which makes it possible to use a mouse instead of keyboard

• GUI is much easier to navigate than CUI

• There is only text in case of CUI whereas there are graphics and other visual clues in case of GUI

• Most modern computers use GUI and not CUI

• DOS is an example of CUI whereas Windows is an example of GUI.

COMPUTER AIDED DESIGN (CAD)

Digital images

Digital images can usually be divided into two distinct categories. They are either **bitmap files** or **vector graphics**.

If you work in a design studio, you need a good understanding on the advantages and disadvantages of both types of data.

These points explain the differences.

- As a general rule digital pictures and scanned images are **bitmap files**. These are sometime also called **raster images.**
- Drawings made in applications like Adobe Illustrator or Corel Draw are saved as **vector graphics.**

Technically both data formats are completely different. The end result however can look virtually identical in either format.

As a general rule bitmaps are typically used to depict **lifelike images** whereas vector graphics are more often used for **abstract images such as logos**.

Bitmap images

Bitmap images are exactly what their name says they are**: a collection of bits that form an image.** The image consists of a matrix of individual dots (or pixels) that all have their own color (described using **bits,** the smallest possible units of information for a computer).

Types of bitmap images

Bitmap images can contain any number of colors but there are four main categories:

1. **Line-art.** These are images that only contain two colors, usually black and white. Sometimes these images are referred to as bitmaps because a computer has to use only 1 bit (on=black, off=white) to define each pixel.

2. **Grayscale images,** which contain various shades of gray as well as pure black and white. Typically 256 shades of gray (8-bit) are used even though the human visual system needs only 100 tints to perceive an image as life-like.

3. **Multitones:** such images contain shades of two or more colors. The most popular multitone images are **duotones,** which usually consist of **black** and a **second spot color.**

4. **Full color images.** The color information can be described using a number of color spaces: RGB, CMYK or Lab for instance. PANTONE

Vector graphics

Vector graphics are images that are completely described using mathematical definitions.

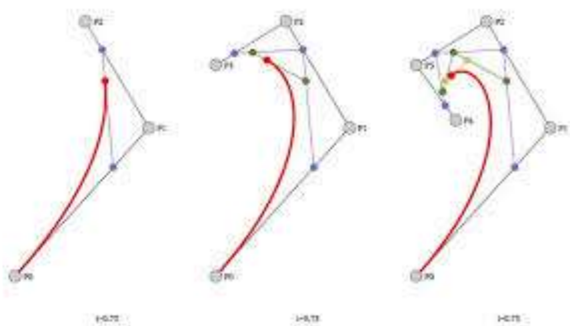
Each individual line is made up of either a vast collection of points with lines interconnecting all of them or just a few control points that are connected using so called Bézier curves. It is this latter method that generates the best results and that is used by most drawing programs.

Characteristics of vector drawings

Vector drawings are usually pretty small files because they only contain data about the Bézier curves that form the drawing.

Bézier curves are **parametric curves** defined by **control points**, widely used in computer graphics, CAD, and vector software (e.g., CorelDRAW, Adobe Illustrator, Inkscape etc) to **create smooth**, **scalable shapes**.

They are defined by a **starting point, ending point,** and **intermediate control points** that shape the curve _ for smooth, flexible lines. etc



Vector drawings can usually be scaled without any loss in quality. This makes them ideal for **company logos**, **maps** or **other objects** that have to be resized frequently.

Note that not all vector drawings can be scaled as much as you like:

- Drawings containing trapping information can only be scaled up to 20 percent larger or smaller.
- **Thin lines may disappear** if a vector drawing is reduced too much.
- **Small errors** in a drawing may become visible as soon as it is enlarged too much.

It is fairly easy to create a vector based drawing.

COMPUTERS AND FASHION

Over the decades computers and fashion have developed gradually, changed with time, taste and trend.

But nobody knew that a time will come when both these fields will complement each other so well.

Today fashion design has reached new heights by computer aided methods of design. As a result of which, computer industry has got its new customer.

Computer technology is making waves in the fashion design zone.

From determining textile weaves to sizing designs; computers are a vital component of the fashion industry.

Computer aided design (CAD) programs reduce the demand for manual sketches.

New software programs continue to replace old manual skills. Going by the wayside are "old fashioned" flat pattern construction, pencil sketching and traditional math-based pattern sizing. Those who lag in math and falter at sketching can now breathe a little easier.

What is CAD?

Computer-aided design (CAD), also known as **computer-aided design and drafting** (CADD), is the use of computer technology for the process of design and design-documentation.

Computer Aided Drafting describes the process of drafting with a computer. CADD software, or environments, provides the user with input-tools for the purpose of streamlining design processes; drafting, documentation, and manufacturing processes.

CAD may be used to **design curves and figures in two-dimensional (2D**) space; or **curves, surfaces, and solids in three-dimensional (3D)** objects.

Although most designers initially sketch designs by hand, a growing number also translate these hand sketches to the computer.

CAD allows designers to view designs of clothing on virtual models and in various colors and shapes, thus saving time by requiring fewer adjustments of prototypes and samples later.

Most fashion design colleges, however, <u>still teach traditional design methods, including manual flat pattern construction, draping and line drawing.</u>

No doubt that learning of these methods are essential for having a good idea about fashion design but Cutting-edge education also focuses on computer aided methods of design.

Software can help **designers draw**, **create woven textures**, **drape models to create patterns**, **adjust sizes** and even **determine fabric colors.**

By Introducing this technological aspect will enable designers to understand a lot better and try various combinations in their design.

This also **cuts down the time factor** i.e. by use of CAD methods designers can learn a lot faster and more software in less time but Fashion Design is not an easy profession.

You don't have to word hard rather WORK SMART.

Blend OLD and NEW

It's not that one should neglect the manual design methods and completely focus on CAD methods.

State-of-the-art technology is important, but a sound understanding of the methods behind production is also essential. (CAM)

Manually figuring size adjustments and cutting pattern pieces instills that knowledge. Software programs constantly evolve.

A program used today may be obsolete within several years.

Being trained on today's software does not guarantee it will be used when you are ready to go out into the field.

Understanding calculations is timeless, as is computer competency. Software, however, shifts rapidly.

CAD Softwares

All designers are expected to be proficient in CAD.

These include:

- Microsoft Office - Word, Excel, PowerPoint, Publisher
- Drawing packages (vector based) – Adobe illustrator, CorelDraw, etc

- Image editing (bitmap based) – Photoshop

Specialist Fashion software, offers a tailor-made solution that improves the design process, bringing elegance to clothes and accessories.

- Tailornova.
- Browzwear.
- Blender.
- CLO 3D software.
- Digital Fashion Pro.
- Vogue Runway.
- Optitex Speed StepValentina

Pros of CAD

Computer aided designs are helpful in that they save both money and time for designers.

Designers who are more technologically advanced often choose to use computers in order to save money.

The use of computers can help to cut down on the costs of production as well a the cost to pay workers.

Without computers, a large amount manpower is needed to both design clothing and produce it.

Using computers to design the clothing, can help designers to save the money and put it towards more quality production of the clothing. Overall, the cost of the computers is well worth the amount of money that I saved in the end.

Additionally, the sharing of clothing is much easier for designers using CADs.

Designers who may want designs reviewed and edited by individuals with different viewpoints can easily share their computer generated designs.

The more traditional method of producing sketches creates a roadblock for such sharing.

Designers who use such methods would have to spend travel time to share their designs and gain feedback.

In addition to sharing designs with other designers, CADs help designers to share sketches with consumers.

The use of computer generated models and designs is quickly growing in marketing.

Cons of CAD – SMALL COTTAGE DESIGN INDUSTRIES

Computer Aided Designs have made the act of physically sketching and drawing a thing of the past.

Is it necessarily a good thing that paper and pencil are becoming obsolete for fashion design? These programs don't do everything for you.

Despite popular belief, many schools still teach the traditional way-drawing and sketching-because math is still required even for basic design.

Measurements are required to make the actual product so hands on work is needed. When it comes time to put the piece of clothing together, the artist must be able to do manual labor.

Even though CADs are easily accessible and popular, they still are not the method of choice for many fashion designers.

As with any technology, Computer Aided Design programs have quirks and problems. Certain designers may not feel like messing with technology and skip CADs all together.

Most programs can be difficult to grasp and learn.

Some artists rather act traditionally and stick to what they know.

CADs are also much more expensive than drawing/drafting supplies.

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29 APRIL 2026

Computer-Aided Manufacturing (CAM)

Definition - What does Computer-Aided Manufacturing (CAM) mean?

Computer-aided manufacturing (CAM) is an application technology that uses computer software and machinery to facilitate and automate manufacturing processes.

CAM also known as Computer-aided Modeling or Computer-aided Machining is the use of software and computer-controlled machinery to automate a manufacturing process.

In addition to materials requirements, modern CAM systems include real-time controls and robotics.

CAM means the use of computer software to control machine in the manufacturing process. CAM is considered as a numerical control programming tool.

The 2D or 3D models of components generated in CAD software are used to generate G-code to drive computer numerically controlled (CNC) machine tools.

CNC Computer-numerical control – TEXTILE INDUSTRY

- Computer-aided manufacturing involves the use of **CNC machines** for printing, cutting, joining and many other textiles processes.
- Conventionally, an operator decides and adjusts various machines parameters like feed, depth of cut etc depending on type of job, and controls the slide movements by hand. In a CNC Machine functions and slide movements are controlled by motors using computer programs.
- CNC-automated machines can repeat processes with accuracy and reliability, and are easily re-programmed when changes to design or production run are needed. The graphic shows some of the uses of CNC machines.

In **textile processing**, these machines are used to :

1. pick up the fabric from store,
2. spread and cut the fabrics,
3. label
4. transport the cut fabric pieces for assembly and to
5. move the cut fabric pieces around the factory on an overhead conveyer.
6. automatic buttonholing
7. automatic embroidery.

(CAM)

CAM reduces waste and energy for enhanced manufacturing and production efficiency via increased production speeds, raw material consistency and more precise tooling accuracy.

CAM uses computer-driven manufacturing processes for additional automation of management, material tracking, planning and transportation.

CAM also implements advanced productivity tools like simulation and optimization to leverage professional skills.

Depending on enterprise solution and manufacturer, CAM may present inadequacies in the following areas:

- Manufacturing process and usage complexity
- Machine process automation

Modern CAM solutions are scalable and range from discrete systems to multi-CAD 3D integration.

CAM is often linked with CAD for more enhanced and streamlined manufacturing, efficient design and superior machinery automation

ICT and computer-aided manufacture (CAM)

ICT and CAM play a vital role in modern textiles production. For example, they enable :

- designs to be sent electronically to the print manufacturer and stored on computer to ease repeat printing orders.
- colours to be matched to the design, dyes weighed and dispensed and the fabric printed automatically
- ICT makes possible the just-in-time ordering of materials and components so they arrive at the factory as they are needed, ie just-in-time for production to start.
- ICT enables companies to transmit information between plants, and manufacture on a global scale

CIM

Computer-integrated manufacturing (CIM) systems integrate or link CAD and CAM systems. These combined systems link design development, production planning and manufacturing systems together.

Computer-Aided Manufacturing (CAM) in the fashion industry uses computer software to automate production processes, transforming digital designs into physical garments.

- It enhances efficiency,
- accuracy,
- sustainability by automating cutting,
- sewing,
- knitting,
- significantly reducing production time, fabric waste, and human error.

Key Aspects of CAM in Fashion:

1. **Automated Cutting:** CAM systems control CNC cutting machines to precisely cut fabric pieces, maximizing material utilization and reducing waste.
2. **Automated Knitting/Sewing:** CAM manages machinery for knitting garments, allowing for fast production of complex patterns and shapes.
3. **Digitization:** It works closely with Computer-Aided Design (CAD), translating digital patterns into machine-ready instructions.
4. **Improved Efficiency:** Real-time data collection optimizes production workflows and minimizes bottlenecks on the factory floor.
5. **Mass Customization:** CAM, often combined with 3D body scanning, enables the efficient production of custom-made clothing.

Benefits:

1. **Speed:** Garment samples can be produced in under an hour, speeding up the design-to-market cycle.
2. **Precision:** High-resolution digital control reduces inaccuracies in cutting and grading.
3. **Cost Efficiency:** While the initial investment is high, CAM reduces waste and labor costs over time.

Challenges:

1. **High Initial Cost:** Implementing CAM systems requires significant investment in technology and staff training.
2. **Technical Skill Requirements:** Operating these systems requires specialized knowledge

Common software and hardware include CAM integrated with CAD programs which allow for seamless CAD-to-CAM workflows.

CAM software examples

- **Lectra/Gerber Technology** (Modaris for 2D/3D patternmaking, Diamino for marker making, and Kubix Link)

- **Optitex** – (*OPTITEX CREATIVE* is an advanced fashion design software offering 2D/3D pattern tools, fabric simulation, and virtual sampling.)
- **Delcam-** (PowerMill**,** PowerSHAPE, PowerINSPECT)
- **Siemens PLM Software (** Teamcenter: A leading PLM platform for managing product data, documentation, and processes, NX: A comprehensive CAD/CAM/CAE solution for 3D design and simulation, Solid Edge: A 3D design system for rapid product development. Tecnomatix: A digital manufacturing software portfolio.)
- **PTC**
- **Vero Software**
- **SolidCAM**
- **CNC Software**
- **BobCAD-CAM**

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